GRADE 12 · TERM 4

Refresher: Grades 10 to 12, Paper by Paper

One last pass through the whole course, organised the way you will be tested.

Name: _____

Class: _____ Date: _____

How to use this workbook

1. Work with a **pencil**. Mistakes are how we learn: rub out and try again.
2. Start each unit with **Think First**. Try it on your own before you read the explanation.
3. Read **Learn It**, then do the activities. Trace tables and code are written **by hand**.
4. At the end of each unit, tick the "I can" boxes you are sure of. Write down anything you still want to ask.
5. The **checkpoint test** is done under test conditions, with no notes.

In this workbook

1. Where do I stand?
2. Programming core: Grade 10 skills
3. Arrays and text files: Grade 11 skills
4. OOP: classes and objects (Grade 12)
5. 2D arrays and recursion (Grade 12)
6. Databases and SQL (Grades 11 and 12)
7. Paper 1 mini-exam: Delphi programming and SQL
8. Hardware and software
9. Networks and the internet
10. Data and databases
11. Number systems and data representation
12. Security, crime and society
13. Exam answering skills
14. Paper 2 mini-exam: Theory, Grades 10 to 12
15. The final stretch

> **Matches the Annual Teaching Plan:** Term 4 — no new content. Revision of Grade 10, 11 and 12 work, Paper 1 then Paper 2, with a mini-exam for each. Every question is tagged G10, G11 or G12 so you can see which year's work needs attention.

UNIT 1 Where do I stand?

You have three years of Information Technology behind you. Before you revise, be honest about where you are strong and where you are shaky, so you spend your time where it counts.

By the end of this unit I can: (tick each box when you are sure)

- ☐ fill in the table below truthfully
- ☐ name my three weakest topics
- ☐ choose where to start my revision
- ☐ make a realistic plan for the weeks ahead

One thing I want to ask my teacher: _____

1.1 Self-audit

Write ✓ if you are confident, **?** if you are nearly there, or ✗ if you are not yet comfortable. The grade shows where the topic was first taught. Come back at the end and change the marks.

Topic	From	My rating
Variables, operators, IF and CASE	G10	
Loops and strings from first principles	G10	
Number algorithms (prime, GCD, LCM)	G10	
1D arrays, searching and sorting	G11	
Text files and parallel arrays	G11	
Procedures and functions	G11	
Databases in Delphi (no SQL)	G11	
OOP: classes and objects	G12	
2D arrays	G12	
Recursion	G12	
SQL and joins	G12	
Hardware, software and the operating system	G10-12	
Networks and the internet	G10-12	
Number systems, ASCII and compression	G10-11	
Data quality, validation and ERDs	G11	
Normalisation and anomalies	G12	
Security, malware and cybercrime	G10-12	
Social implications: 4IR, IoT, Big Data, AI, cloud	G11-12	
Writing exam answers in paragraphs	All	

1.2 My plan

Write the **three weakest topics** from your table, and for each one the **first thing you will do** this week (for example: redo the trace tables on paper, or write the normalisation steps again).

UNIT 2 Programming core: Grade 10 skills

Everything later in the course is built on four old friends: variables and operators, decisions, loops and strings. Warm up here, because Question 1 of the practical paper tests them every year.

By the end of this unit I can: (tick each box when you are sure)

- ☐ evaluate expressions with div, mod, Round, Trunc and string functions
- ☐ trace IF, CASE and all three loops by hand
- ☐ pull a string apart with Pos, Copy and Delete
- ☐ write number algorithms: digits, factors and primes
- ☐ find and fix the classic loop and decision errors

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

2.1 Counting the change

A till must hand back R137 using only R50 notes, R20 notes and R10 notes, and then coins for whatever is left. Work out by hand how many of each note, and how much is left for coins. Then say which two Delphi operators did the work in each step.

LEARN IT

Operators and decisions: the rules and the traps

- **/** always gives a Real (*23 / 4 = 5.75*); **div** and **mod** work only on whole numbers. **Trunc** chops the decimals off, **Round** goes to the nearest whole number.
- Order of work: brackets, then **not**, then *** / div mod and**, then **+ - or**. Put brackets round every comparison: *(iAge > 12) and (iAge < 19)*.
- Never put a **;** before **else**. Use **begin…end** when a branch has more than one statement.
- **CASE** needs a whole-number or Char selector (not a string or Real). Ranges are written *80..100*, and **else** catches everything left.
- Text comparisons are case-sensitive ('a' > 'A'): use **UpperCase** before comparing user input. **Pos** gives 0 when nothing is found.

Delphi

```
case iMark of
  80..100 : sSym := 'A';
  60..79  : sSym := 'B';
  else      sSym := 'C';
end;
if (iMark mod 10 = 0) or (iMark = 99) then   // brackets round each test
  sSym := sSym + '*';
```

LEARN IT

Loops, strings and number algorithms: the patterns

- **FOR** when you know how many times; **WHILE** (test first, may run 0 times) and **REPEAT…UNTIL** (test last, runs at least once, stops when the test is True) when you do not. Read the next value **inside** the loop, or it never ends.
- Total starts at 0, counter starts at 0 and a flag starts True (assume prime) or False (assume not found) **before** the loop.
- Digits of a whole number: *mod 10* gives the last digit, *div 10* removes it.
- Factors: *iNum mod i = 0*. A prime has exactly two factors. The GCD is found by repeating *remainder* until it is 0.

- Strings start at position 1. *Copy(s, start, count)* returns text; *Delete(s, start, count)* and *Insert* are procedures that change s. Build a new string with *s := s + ch*.

Delphi

```
while iB <> 0 do            // GCD by remainders
begin
  iTemp := iB;
  iB := iA mod iB;
  iA := iTemp;
end;                         // answer is in iA

iSp := Pos(' ', sName);
sFirst := Copy(sName, 1, iSp - 1);
Delete(sName, 1, iSp);       // sName is now the rest
```

2.2 [G10] Evaluate [8]

Write the value of each expression. Real values are written as Delphi would show them.

Expression	Value
23 div 4	
23 mod 4	
23 / 4	
Round(6.7) + Trunc(6.7)	
Sqr(5) - Sqrt(16)	
(23 mod 4 = 3) and (23 div 4 > 5)	
Length('Cape Town') - Pos('T', 'Cape Town')	
Chr(Ord('A') + 7)	

2.3 [G10] CASE and a bonus [7]

Write the symbol stored in **sSym** at the end of the code for each mark.

Delphi

```
case iMark of
  80..100 : sSym := 'A';
  70..79  : sSym := 'B';
  60..69  : sSym := 'C';
  50..59  : sSym := 'D';
  else      sSym := 'F';
end;
if (iMark mod 10 >= 8) and (sSym <> 'F') then
  sSym := sSym + '+';
```

iMark	sSym at the end
85	
59	
70	
49	
100	
79	
88	

2.4 [G10] Digits with a WHILE loop [6]

Trace the code with iNum = 4093 at the start. Write the values **after** each pass of the loop.

Delphi

```
iRev := 0;
iSum := 0;
while iNum > 0 do
begin
  iDigit := iNum mod 10;
  iRev := iRev * 10 + iDigit;
  iSum := iSum + iDigit;
  iNum := iNum div 10;
end;
```

Pass	iDigit	iRev	iSum	iNum
1				
2				
3				
4				

2.5 [G10] Debug: is it prime? [4]

This code should show *Prime* in **lblOut** when **iNum** is a prime number, but it is wrong for every number. Find **four** errors.

Delphi

```
bPrime := False;
for i := 2 to iNum do
  if iNum div i = 0 then
    bPrime := True;
if bPrime then
  lblOut.Caption := 'Prime';
```

2.6 [G10] Write it: read an ID number [7]

A South African ID number is 13 digits: **YYMMDD** (birth date), then **SSSS** (4 digits: 5000 or more is male, below 5000 is female), then a citizenship digit (position 11: 0 is a citizen, 1 a permanent resident). **sID** holds the number and every learner was born in the 2000s. Write code that shows in **lblInfo** the birth date as *dd/mm/yyyy*, the gender and the citizenship, for example *02/04/2009, Male, South African citizen*.

UNIT 3 Arrays and text files: Grade 11 skills

Arrays hold a list in memory and text files keep it after the program closes. Put them together with procedures and functions and you can load, search, sort and save data. This unit revises exactly that chain.

By the end of this unit I can: (tick each box when you are sure)

- ☐ use the standard array patterns: total, highest with its index, search and sort
- ☐ trace a bubble sort and a binary search pass by pass
- ☐ read a text file line by line and split each line
- ☐ write and append to text files without erasing them by accident
- ☐ write functions and procedures with value and reference parameters

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

3.1 The big phone book

A sorted list has **15** surnames. You look at the middle name; it tells you whether your surname comes before or after it, so you can throw half the list away. (a) How many names are left after your first look? After the second? After the third? (b) What is the largest number of looks you will ever need for 15 names? (c) A list of **1 000** sorted names: is the answer closer to 10 or to 500?

LEARN IT

Arrays: the patterns to know by heart

- **Total and average:** total starts at 0; average = total / count (a Real).
- **Highest or lowest with its place:** start with item 1 as the best; store the **index** (not only the value) when you find a better one.
- **Linear search:** a flag and a WHILE loop that stops when found or when the end is reached. It works on any list.
- **Bubble sort:** compare neighbours and swap with a temporary variable. Pass i needs only *n* - *i* comparisons. With **parallel arrays**, swap **every** array or the names and marks drift apart.
- **Binary search:** only on a **sorted** list. Compare the middle item, then move iLow or iHigh past it.

Delphi

```
iBest := 1;                          // highest with index
for i := 2 to 8 do
  if arrTimes[i] > arrTimes[iBest] then
    iBest := i;

for i := 1 to n - 1 do               // bubble sort, ascending
  for j := 1 to n - i do
    if arrMark[j] > arrMark[j + 1] then
    begin
      iTemp := arrMark[j];  arrMark[j] := arrMark[j + 1];  arrMark[j + 1] := iTemp;
      sTemp := arrName[j];  arrName[j] := arrName[j + 1];  arrName[j + 1] := sTemp;
    end;
```

LEARN IT

Text files, procedures and functions

- **Order:** AssignFile, then Reset (read), Rewrite (write: **erases** the old contents) or Append (write at the end), then the loop, then **CloseFile**. Use *FileExists* before Reset.

- Read with *while not Eof(tFile) do ReadLn(tFile, sLine)*. Split a line with Pos, Copy and Delete. Write with *WriteLn(tFile, text)*. A file must be closed before it is reopened.
- A **function** returns one value through *Result*; a **procedure** does an action. Parameters are copies (**value** parameters) unless marked **var**: only *var* parameters send a changed value back to the caller.
- Pass an array by declaring a type first: *type TTimes = array[1..8] of Integer;*

Delphi

```
AssignFile(tFile, 'names.txt');
Reset(tFile);
while not Eof(tFile) do
begin
  ReadLn(tFile, sLine);
  memOut.Lines.Add(sLine);
end;
CloseFile(tFile);

procedure Swap(var iA, iB: Integer);    // var: the caller's values change
var iTemp: Integer;
begin
  iTemp := iA;  iA := iB;  iB := iTemp;
end;
```

3.2 [G11] Bubble sort pass by pass [7]

Sort **41, 17, 33, 9, 25, 12** into ascending order with a bubble sort (the code in Learn It, n = 6). Write the whole list after each pass and the number of swaps made in that pass.

Pass	List after the pass	Swaps
1		
2		
3		
4		
5		

3.3 [G11] Binary search by hand [7]

The sorted array **arrCodes** holds 102, 115, 121, 134, 148, 152, 167, 173, 189, 195, 204 (positions 1 to 11). Trace the search for **iTarget = 160**.

Delphi

```
iLow := 1;  iHigh := 11;  bFound := False;
while (iLow <= iHigh) and (not bFound) do
begin
  iMid := (iLow + iHigh) div 2;
  if arrCodes[iMid] = iTarget then
    bFound := True
  else if arrCodes[iMid] < iTarget then
    iLow := iMid + 1
  else
    iHigh := iMid - 1;
end;
```

Pass	iLow	iHigh	iMid	arrCodes[iMid]	What happens
1					
2					
3					
end					

3.4 [G11] Trace a text file [8]

The file *stock.txt* has the five lines below (name#price#quantity). Complete the table for the code. **iTotal** is the running total of price × quantity.

stock.txt

```
Maize meal#43#18
Sugar 2kg#39#7
Cooking oil#54#12
Rusks#31#4
Tea bags#27#20
```

Delphi

```
AssignFile(tFile, 'stock.txt');
Reset(tFile);
iTotal := 0;
while not Eof(tFile) do
begin
  ReadLn(tFile, sLine);
  iPos := Pos('#', sLine);
  sName := Copy(sLine, 1, iPos - 1);
  Delete(sLine, 1, iPos);
  iPos := Pos('#', sLine);
  iPrice := StrToInt(Copy(sLine, 1, iPos - 1));
  iQty := StrToInt(Copy(sLine, iPos + 1, Length(sLine)));
  iTotal := iTotal + iPrice * iQty;
  if iQty < 10 then
    memLow.Lines.Add(sName);
end;
CloseFile(tFile);
```

Line	sName	iPrice	iQty	iTotal after	Added to memLow?
1					
2					
3					
4					
5					

3.5 [G11] Debug: add to a log file [5]

The code should **add** the text in edtEntry to the **end** of the existing file *log.txt* and then show in lblLines how many lines the file now has. Find **five** errors.

Delphi

```
AssignFile(tLog, 'log.txt');
Rewrite(tLog);
WriteLn(tLog, edtEntry.Text);
Reset(tLog);
iLines := 0;
while Eof(tLog) do
  ReadLn(tLog, sLine);
lblLines.Caption := IntToStr(iLines);
```

3.6 **[G11] Write it: a procedure with var parameters** **[8]**

Declare *type TTimes = array[1..8] of Integer;* and write a **procedure MinMax** that receives a TTimes array and sends back the lowest and highest times through **var** parameters *iMin* and *iMax*. Then write the call that stores them in **iFast** and **iLast**.

UNIT 4 OOP: classes and objects (Grade 12)

A class is a blueprint and an object is one thing built from it. Question 3 of the practical paper always asks for the same pattern: attributes, a constructor, accessors, a mutator, an auxiliary method and ToString, then code that uses the objects.

By the end of this unit I can: (tick each box when you are sure)

- ☐ list the attributes and methods a class needs from a short story
- ☐ say what constructor, accessor, mutator, auxiliary and ToString mean
- ☐ trace an object's attributes call by call
- ☐ write the methods of a class and use the objects in a form
- ☐ explain what happens when one object variable is assigned to another

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

4.1 Blueprint for an airtime wallet

A learner uses a phone app that keeps an airtime wallet for one person. The wallet knows its **owner's name** and its **balance in rand**. The owner can **top up** (only with an amount above 0) and can **spend** money. (a) List the private attributes and their data types. (b) List the public methods. (c) What should *Spend* do if the balance is too small, and what should it give back to the caller? (d) Why must the balance be private?

LEARN IT

A complete class: TPlayer (netball shooting record)

Private attributes (prefix f); **public** methods. The **constructor** sets the starting values. An **accessor** (Get) returns an attribute. A **mutator** (Set, AddShot) changes one, with a rule if needed. An **auxiliary** method is a private helper that works something out and never changes an attribute. **ToString** returns the object's state as text.

Delphi

```
type
  TPlayer = class
  private
    fName     : string;
    fGoals    : Integer;
    fAttempts : Integer;
    function Accuracy : Real;              // auxiliary: private helper
  public
    constructor Create(sName: string);
    function    GetGoals : Integer;        // accessor
    procedure   SetName(sNew: string);     // mutator
    procedure   AddShot(bScored: Boolean); // mutator
    function    ToString : string;
  end;

constructor TPlayer.Create(sName: string);
begin
  fName := sName;  fGoals := 0;  fAttempts := 0;
end;

function TPlayer.GetGoals : Integer;
begin
  Result := fGoals;
```

```
end;

procedure TPlayer.SetName(sNew: string);
begin
  if sNew <> '' then                        // the mutator protects the attribute
    fName := sNew;
end;

procedure TPlayer.AddShot(bScored: Boolean);
begin
  Inc(fAttempts);
  if bScored then Inc(fGoals);
end;

function TPlayer.Accuracy : Real;
begin
  if fAttempts = 0 then Result := 0
  else Result := fGoals / fAttempts * 100;
end;

function TPlayer.ToString : string;
begin
  Result := fName + ': ' + IntToStr(fGoals) + '/' + IntToStr(fAttempts)
            + ' (' + FloatToStrF(Accuracy, ffFixed, 5, 1) + '%)';
end;
```

LEARN IT

The class used in the next questions: TTrip

A minibus taxi route stores its **route name**, its **capacity**, how many seats are **booked** and the **fare per seat**. **Book** adds seats only if enough are left and says whether it worked. **SeatsLeft** is a private helper. **IsFull** is True when no seats are left. **Income** is booked × fare.

Delphi

```
type
  TTrip = class
  private
    fRoute    : string;
    fCapacity : Integer;
    fBooked   : Integer;
    fFare     : Real;
    function SeatsLeft : Integer;
  public
    constructor Create(sRoute: string; iCapacity: Integer; rFare: Real);
    function    GetBooked : Integer;
    procedure   SetFare(rNew: Real);
    function    Book(iSeats: Integer) : Boolean;
    function    Income : Real;
    function    IsFull : Boolean;
    function    ToString : string;
  end;
```

4.2 [G12] Be the computer: an object and its alias [7]

Complete the table for the object **objT** after each statement. *SetFare* accepts only a fare above 0, and *Book* accepts a booking only when enough seats are left. Show the money with 2 decimals.

Delphi

```
objT := TTrip.Create('Bellville-CBD', 15, 24.50);
objT.Book(6);
objT.Book(10);
objT.SetFare(-5);
objU := objT;
objU.SetFare(26);
objU.Book(9);
```

After this statement	fBooked	fFare (R)	Book returned	Income (R)
objT := TTrip.Create('Bellville-CBD', 15, 24.50);				
objT.Book(6);				
objT.Book(10);				
objT.SetFare(-5);				
objU := objT;				
objU.SetFare(26);				
objU.Book(9);				

4.3 [G12] Debug: a faulty class [4]

This part of TTrip compiles, but the program gives wrong answers. Find **four** errors.

Delphi

```
constructor TTrip.Create(sRoute: string; iCapacity: Integer; rFare: Real);
begin
  fRoute := sRoute;
  iCapacity := fCapacity;
  fFare := rFare;
end;

function TTrip.Book(iSeats: Integer) : Boolean;
begin
  if iSeats <= SeatsLeft then
    fBooked := iSeats;
  Result := True;
end;
```

4.4 [G12] Write it: the constructor and two methods [7]

Use the TTrip interface above. Write the implementation of: the **constructor** (no seats are booked at the start); the mutator **SetFare** (accepts only a fare above 0); and **Book** (adds the seats and returns True **only if enough seats are left**, otherwise nothing changes and it returns False).

4.5 [G12] Write it: an array of objects in the form **[8]**

The five trips below have already been created and stored in **arrTrips[1..5]** (array of TTrip), with the bookings shown:

	Route	Capacity	Booked	Fare (R)
1	Bellville-CBD	15	15	24.50
2	Parow-Century City	12	9	28.00
3	Durbanville-Bellville	15	11	19.50
4	Goodwood-CBD	10	10	22.00
5	Kuils River-Bellville	14	6	17.00

Write the code to (a) show the total income of all five trips in **lblTotal** with 2 decimals; (b) count the trips that are **full** and show the number in **lblFull**; and (c) add the **ToString** of the trip with the **highest income** to **memOut**.

UNIT 5 2D arrays and recursion (Grade 12)

A 2D array is a table in memory: the first index chooses the row and the second chooses the column. Recursion solves a problem by calling itself on a smaller one. Both give the practical paper a short question every year.

By the end of this unit I can: (tick each box when you are sure)

- ☐ decide which index stays fixed when totalling a row, a column or a diagonal
- ☐ write and trace nested loops over a 2D array
- ☐ explain base case and recursive case in my own words
- ☐ trace a recursive function call by call and then back up
- ☐ write a short recursive function

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

5.1 The taxi-rank grid

The number of people waiting at four taxi ranks was counted each hour from 06:00 to 09:00. Complete the totals by hand.

Rank	06:00	07:00	08:00	09:00	Total
Bellville	12	25	31	18	
Parow	8	20	27	22	
Goodwood	15	30	38	26	
Durbanville	5	12	16	10	
Hour total					

(a) Which hour was the busiest overall? (b) Add the four cells that run from the top-left corner to the bottom-right corner. (c) To total **Goodwood's** row, which number stays the same as you move from cell to cell, and which changes?

THINK FIRST · no computer needed

5.2 Folding paper

A sheet of paper is folded in half again and again. Layers(0) = 1 (the flat sheet). Write the number of layers after 1, 2, 3 and 4 folds: _____, _____, _____, _____. Then complete: Layers(n) = _____ × Layers(_____). What is Layers(6)? Which value is known without any further folding?

LEARN IT

2D arrays: which index stays fixed?

arrQ[r, c]: r is the row, c the column. **Row total:** fix r, loop c. **Column total:** fix c, loop r. **Main diagonal:** r and c are equal. Reset a total to 0 **before** the loop that builds it (inside the outer loop when there is one total per row or column). To remember **where** the best value is, keep the row or column number, not only the value. Build one string per row with the inner loop and add it to the memo after the inner loop ends.

Delphi

```
iRow3 := 0;                       // row total: row fixed, column changes
for c := 1 to 4 do
  iRow3 := iRow3 + arrQ[3, c];

for r := 1 to 4 do                // total of every row
begin
  iT := 0;
  for c := 1 to 4 do
    iT := iT + arrQ[r, c];
  memOut.Lines.Add(IntToStr(iT));
end;

iDiag := 0;                       // diagonal: one loop, r = c
for i := 1 to 4 do
  iDiag := iDiag + arrQ[i, i];
```

LEARN IT

Recursion: a function that calls itself

Every recursive routine needs a **base case** (an answer that needs no further call) and a **recursive case** that calls itself with a **smaller** problem, moving towards the base case. Without a base case the calls never stop (a stack overflow). To trace it, write down each call until the base case; then fill in the returned values from the **bottom upwards**. Anything written after the recursive call happens on the way back.

Delphi

```
function Factorial(n : Integer) : Integer;
begin
  if n <= 1 then
    Result := 1                       // base case
  else
    Result := n * Factorial(n - 1);   // recursive case
end;
// Factorial(4) = 4 * Factorial(3) = 4 * (3 * Factorial(2)) = 4 * 3 * 2 * 1 = 24
```

5.3 [G10] Predict: nested loops [4]

Write what is added to **memOut**. (The inner loop builds one line; *Trim* removes the space at the end.)

Delphi

```
for i := 1 to 4 do
begin
  sLine := '';
  for j := 1 to i do
    sLine := sLine + IntToStr(i * j) + ' ';
  memOut.Lines.Add(Trim(sLine));
end;
```

5.4 [G12] Busiest hour [4]

Use the taxi-rank grid in Think First (**arrQ[rank, hour]**). Write the values after each pass of the outer loop.

```delphi
iBest := 1;  iBestTot := 0;
for c := 1 to 4 do
begin
  iColTot := 0;
  for r := 1 to 4 do
    iColTot := iColTot + arrQ[r, c];
  if iColTot > iBestTot then
  begin
    iBestTot := iColTot;
    iBest := c;
  end;
end;
```

c	iColTot	iBestTot	iBest
1			
2			
3			
4			

5.5 [G12] Write it: the grid questions [6]

Use **arrQ[1..4, 1..4]** from Think First. Write code to: (a) store the total for **Goodwood** (row 3) in **iRow3**; (b) count in **iQuiet** the cells with a queue of **fewer than 10**; (c) store the sum of the **main diagonal** in **iDiag**.

5.6 [G12] Trace a recursive function [5]

Trace **Mix(5)**. List every call, say whether the base case is reached, and write the value each call **returns** (fill in from the bottom row upwards).

Delphi

```
function Mix(n : Integer) : Integer;
begin
  if n <= 1 then
    Result := 2
  else
    Result := 3 * Mix(n - 1) - n;
end;
```

Call	n	n <= 1 ?	Value returned
Mix(5)	5		
Mix(4)	4		
Mix(3)	3		
Mix(2)	2		
Mix(1)	1		

5.7 [G12] Predict: before and after the call [3]

Write the lines added to **memOut** by **Echo(3)**. Notice what is printed on the way *down* and on the way *back*.

Delphi

```
procedure Echo(n : Integer);
begin
  if n > 0 then
  begin
    memOut.Lines.Add('in ' + IntToStr(n));
    Echo(n - 1);
    memOut.Lines.Add('out ' + IntToStr(n));
  end;
end;
```

5.8 [G12] Write it: two recursive functions [6]

(a) Write **function Gcd(a, b : Integer) : Integer**: if b is 0 the answer is a, otherwise it is the Gcd of b and *a mod b*.
(b) Write **function Rev(s : string) : string** that returns the text backwards (*Table* gives *elbaT*): an empty string is its own reverse; otherwise it is the reverse of all but the first character, followed by the first character.

UNIT 6 Databases and SQL (Grades 11 and 12)

In Grade 11 you handled one table with Delphi code only. In Grade 12 you let SQL do the work, join two tables, and build the SQL inside Delphi. This unit uses one small database: the results of a school sports day.

By the end of this unit I can: (tick each box when you are sure)

- ☐ explain primary key, foreign key and why data is split over two tables
- ☐ read and write Delphi code that loops through a table without SQL
- ☐ write SELECT, WHERE, ORDER BY, TOP and aggregate queries
- ☐ write a join with GROUP BY and HAVING and predict its result
- ☐ write INSERT, UPDATE and DELETE and build SQL in Delphi with Open and ExecSQL

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

6.1 One table or two?

A teacher keeps every athlete in one big table, repeating the house and its captain on every row:

Athlete	House	House captain
Thabo Mahlangu	Impala	Lerato Sithole
Ruan Pretorius	Impala	Lerato Sithole
Mpho Sekhukhune	Impala	Lerato Sithole
Aisha Cassiem	Kudu	Ruan Visser

(a) Impala gets a new captain. How many rows must be changed, and what could go wrong? (b) How would you split the data into **two tables**? Name the table that holds the captain, and say which field links the two tables.

LEARN IT

The database used in this unit

tblHouse (primary key HouseID) and **tblAthlete** (primary key AthleteID; HouseID is the foreign key). Dates are Date/Time fields.

HouseID	HouseName	Colour	Captain
1	Impala	Gold	Lerato Sithole
2	Kudu	Green	Ruan Visser
3	Eland	Blue	Nomsa Zulu
4	Springbok	Red	Kiara Naidoo

ID	Name	Grade	HouseID	Event	Points	EventDate
1	Thabo Mahlangu	12	1	100m	8	2026/02/20
2	Aisha Cassiem	11	2	100m	6	2026/02/20
3	Pieter Smit	12	3	Long jump	9	2026/03/04
4	Zinhle Dube	10	4	100m	4	2026/02/20

5	Ruan Pretorius	11	1	Relay	7	2026/03/04
6	Naledi Khumalo	12	2	Relay	10	2026/03/11
7	Cameron Adams	10	3	Long jump	5	2026/03/04
8	Lindiwe Ncube	12	4	Long jump	8	2026/03/04
9	Jaden Petersen	11	3	Relay	6	2026/03/11
10	Mpho Sekhukhune	12	1	Long jump	3	2026/03/04
11	Keegan Louw	10	2	Relay	0	2026/03/11
12	Amahle Zondi	11	4	100m	7	2026/02/20

LEARN IT

Database code: Delphi tables, SQL and Delphi queries

Grade 11: a *TADOTable* is read with *First, Eof, Next*; a field is *tblAthlete['Points']*; *Edit* or *Insert* then *Post* changes the table. *Delete* removes the current record and moves to the next one by itself, so do not call Next after it. Never use Locate. **Grade 12:** SQL runs in a *TADOQuery*: *Close*, set *SQL.Text*, then *Open* for a SELECT or *ExecSQL* for INSERT, UPDATE and DELETE. Text from the user goes inside quotes, numbers are joined with IntToStr.

Delphi

```
tblAthlete.First;                          // G11: a TADOTable, no SQL
while not tblAthlete.Eof do
begin
  if tblAthlete['Points'] >= 8 then
  begin
    tblAthlete.Edit;                       // Edit ... Post to change a record
    tblAthlete['Points'] := tblAthlete['Points'] + 1;
    tblAthlete.Post;
  end;
  tblAthlete.Next;                         // never forget Next
end;
tblAthlete.Insert;                         // Insert ... Post adds a record
tblAthlete['Name'] := 'Sizwe Mkhize';
tblAthlete.Post;

qryAthlete.Close;                          // G12: a TADOQuery and SQL
qryAthlete.SQL.Text := 'SELECT Name FROM tblAthlete WHERE Grade = ' + IntToStr(sedGrade.Value);
qryAthlete.Open;                           // SELECT: Open
qryAthlete.SQL.Text := 'DELETE FROM tblAthlete WHERE Points = 0';
qryAthlete.ExecSQL;                        // INSERT, UPDATE, DELETE: ExecSQL
```

LEARN IT

SQL: clause order, joins and the traps

The computer works in this order: **FROM** (join the tables), **WHERE** (keep records), **GROUP BY**, **HAVING** (keep groups), **SELECT**, **ORDER BY**.

- Aggregate functions (COUNT, SUM, AVG, MIN, MAX) never go in WHERE: use **HAVING**. Every other field in SELECT must be in GROUP BY.
- To join two tables, name both after FROM and put the **join condition first in WHERE** (primary key = foreign key); add any filters with AND. Without it every row is paired with every row (a cartesian product).
- Text in double quotes, dates between # signs as #yyyy/mm/dd#, *IS NULL* (never = NULL), *LIKE* with % as the wildcard. AND is worked out before OR: use brackets.
- UPDATE and DELETE without WHERE change **every** record.

SQL

```
SELECT   Event, COUNT(*) AS Entries, AVG(Points) AS AvgPts
FROM     tblAthlete, tblHouse
WHERE    tblAthlete.HouseID = tblHouse.HouseID
         AND EventDate >= #2026/03/01# AND Name LIKE "A%"
GROUP BY Event
```

```
HAVING   COUNT(*) > 1
ORDER BY Entries DESC;

INSERT INTO tblHouse (HouseName, Colour) VALUES ("Oryx", "Purple");
UPDATE tblAthlete SET Points = Points + 1 WHERE Event = "Relay";
DELETE FROM tblAthlete WHERE Points = 0;
```

6.2 [G11] Predict: looping through a table [3]

The table tblAthlete holds the records in the database above (in the order shown). Write what is added to **memOut** and what **lblAvg** shows.

Delphi

```
tblAthlete.First;
iCount := 0;
iTotal := 0;
while not tblAthlete.Eof do
begin
  if tblAthlete['Grade'] = 11 then
  begin
    Inc(iCount);
    iTotal := iTotal + tblAthlete['Points'];
    memOut.Lines.Add(tblAthlete['Name']);
  end;
  tblAthlete.Next;
end;
lblAvg.Caption := FloatToStrF(iTotal / iCount, ffFixed, 5, 1);
```

6.3 [G11] Write it: change a table without SQL [8]

Use the TADOTable **tblAthlete** (no SQL). (a) Add **2 points** to every **Grade 12** athlete whose event is **Relay**. (b) Add a new record: name from **edtName**, grade 10, house 2, event 100m, 0 points. (c) Delete every athlete with **0 points**.

6.4 [G12] Result of a query: AND and OR [4]

A learner wrote the query below. Write the result in the table (as many rows as there are results; leave extra rows empty).

SQL

```
SELECT Name
FROM tblAthlete
WHERE Event = "100m" OR Event = "Relay" AND Points >= 7
ORDER BY Name;
```

Name

6.5 [G12] Result of a query: join, GROUP BY and HAVING [4]

Write the result of this query on the two tables (it joins the tables, adds up each house's points and keeps only the houses with more than 18 points).

SQL

```
SELECT HouseName, COUNT(*) AS Entries, SUM(Points) AS Total
FROM tblHouse, tblAthlete
WHERE tblHouse.HouseID = tblAthlete.HouseID
GROUP BY HouseName
HAVING SUM(Points) > 18
ORDER BY SUM(Points) DESC;
```

HouseName	Entries	Total

6.6 [G12] Write it: six SQL statements [13]

Write the SQL for each: (a) the **two** athletes with the highest points (name and points); (b) the number of athletes and the average points for each event, with the headings Entries and AvgPoints; (c) the name and house name of every athlete whose EventDate is on or after 4 March 2026, ordered by house name and then name; (d) add athlete *Sizwe Mkhize*, Grade 10, house 2, event 100m, 5 points, date 18 March 2026; (e) add 2 points to every Relay athlete in house 3; (f) delete every athlete with 0 points.

6.7 [G12] Debug: two faulty queries [4]

Both queries should work on the sports-day tables. **Query 1** should list each house with the number of its athletes, for houses with more than 2 athletes. **Query 2** should show the name and points of the **3** best athletes since 4 March 2026. Find **four** errors in total.

SQL

```
SELECT HouseName, COUNT(*) AS Entries
FROM tblHouse, tblAathlete
WHERE COUNT(*) > 2
GROUP BY HouseName;

SELECT Name, Points
FROM tblAthlete
WHERE EventDate > 2026/03/04
ORDER BY Points DESC
TOP 3;
```

6.8 [G12] Write it: build the SQL in Delphi [9]

qryAthlete is a TADOQuery. (a) The user picks an event in **cmbEvent** and types a minimum in **sedMin**. List in **memOut** the name, house name and points of each athlete in that event with at least that many points, highest points first, as *Name (House): Points*, and show the number found in **lblCount**. (b) In **btnBonus**, add the number in **sedBonus** to the points of every athlete in the event chosen in cmbEvent.

PAPER 1 MINI-EXAM

Delphi programming and SQL

Time: 120 minutes **Total:** 78 marks **Mark:** _____ / 78

No notes, no computer. Show all your working. Write Delphi code and SQL neatly, with correct indentation.

SECTION A — General programming: strings, loops, decisions and recursion [20]

A1 [6]

A school generates each learner's computer username from the full name typed in **edtName** (for example *Thabo Molefe*) and the birth year in **sedYear**. The username is the **first letter** of the first name, then the whole **surname**, both in small letters, followed by the **last two digits** of the year (with a leading zero if needed). For *Thabo Molefe* born in 2009 the username is *tmolefe09*; born in 2005 it would end in *05*. Write code that shows the username in **lblUser**. The name always has exactly one space.

A2 **[4]**

Trace the code with **iN = 6** at the start. Write the values **after** each pass of the loop.

Delphi

```
iSteps := 0;
while iN <> 1 do
begin
  if Odd(iN) then
    iN := 3 * iN + 1
  else
    iN := iN div 2;
  Inc(iSteps);
end;
```

Pass	iN	iSteps
1		
2		
3		
4		
5		
6		
7		
8		

A3 **[5]**

A weather station's program reads temperatures with an InputBox until the user types the word **stop** (in capitals or small letters). Show the **highest** temperature in **lblHigh** and the **average** to one decimal in **lblAvg**. If no temperature was entered, show nothing. The word *stop* must not be treated as a temperature.

A4 **[5]**

The function **CountSevens** returns how many times the digit **7** appears in a whole number, using **recursion**: for 0 the answer is 0; otherwise it looks at the last digit (*mod 10*), adds 1 if it is a 7, and uses the function again on the number without its last digit (*div 10*). (a) Write the function. (b) State the value returned by *CountSevens(70772)* and the number of times the function is called altogether.

SECTION B — Arrays and text files (with a 2D array) [20]

B1 **[5]**

The text file *tickets.txt* lists matric-dance ticket sales: **surname#grade#tickets** on every line (at most 30 lines).

tickets.txt

```
Dlamini#12#4
Pieterse#11#2
Mokoena#12#6
Abrahams#10#1
Naidoo#12#3
Venter#11#5
Zulu#10#2
```

Write code that reads the file into the parallel arrays **arrSurname**, **arrGrade** and **arrTickets** (the grade and tickets as integers) and counts the lines in **iCount**.

B2 **[3]**

The user types a surname in **edtFind**. Search the arrays (use a **linear search**) and show *Grade 12: 4 tickets* (the learner's own grade and tickets) in **lblFound**, or *Not found*.

B3 **[5]**

Sort the three parallel arrays into **descending order of tickets** with a bubble sort, so that the learner with the most tickets is first.

B4 **[3]**

After the sort, write every learner with **4 or more tickets** to a **new** text file *big.txt*, one per line, as *Surname: tickets* (for example *Mokoena: 6*).

B5 **[4]**

The sales of tickets by grade in four weeks are stored in **arrSales[1..3, 1..4]** (row = grade 10, 11, 12; column = week).

	Week 1	Week 2	Week 3	Week 4
Grade 10	12	18	9	15

Grade 11	20	14	22	19
Grade 12	25	30	28	35

Write code to (a) store the Grade 12 total in **iG12**; (b) find the **week number** with the highest total over all three grades and store it in **iBestWeek**.

SECTION C — Object-oriented programming [18]

C1 [3]

A library records every book that a learner has out. A book may be kept for 14 days; every extra day costs a fine of R2.50. The class **TBookLoan** is declared below.

```delphi
type
  TBookLoan = class
  private
    fTitle : string;
    fDays  : Integer;              // days the book has been out
    function DaysLate : Integer;
  public
    constructor Create(sTitle: string; iDays: Integer);
    function    GetTitle : string;
    procedure   SetDays(iNew: Integer);
    function    Fine : Real;
    function    ToString : string;
  end;
```

Write the type of method from the word bank. *Word bank: constructor · accessor · mutator · auxiliary · ToString*

Method	Type
Create	
SetDays	
DaysLate	

C2 **[4]**

Write the implementation of the **constructor** and of the mutator **SetDays**, which accepts a new number of days only if it is **0 or more**.

C3 **[4]**

Write the private auxiliary function **DaysLate** (the days beyond 14, or 0 if the book is not late) and the function **Fine**, which uses DaysLate and returns the fine in rand.

C4 **[3]**

Write **ToString**, which returns text such as *Long Walk to Freedom: 31 days, fine R42.50* (the fine always with 2 decimals).

C5 **[4]**

Five loans were created and stored in **arrLoans[1..5]** (array of TBookLoan):

	Title	Days out
1	Born a Crime	14
2	Spud	9
3	Long Walk to Freedom	31
4	Cry, the Beloved Country	20
5	Coconut	15

Write code that (a) shows the **total of all the fines** in **lblTotal** with 2 decimals and (b) adds the ToString of every loan that has a **fine above 0** to **memOut**.

SECTION D — Databases and SQL [20]

D1 [4]

A food bank keeps its donors and their donations in two tables:

Table	Fields
tblDonor	DonorID (primary key), DonorName, Town
tblDonation	DonationID (primary key), DonorID (foreign key), Item, Qty, DonationDate

DonationID	DonorID	Item	Qty	DonationDate
1	1	Maize meal	40	2026/05/04
2	2	Canned beans	24	2026/05/04
3	3	Soap	60	2026/05/11
4	1	Canned beans	36	2026/05/18
5	4	Maize meal	15	2026/05/18
6	2	Soap	12	2026/06/01
7	3	Maize meal	30	2026/06/01
8	1	Soap	25	2026/06/08
9	4	Canned beans	18	2026/06/08
10	3	Canned beans	48	2026/06/15

DonorID 1 is Bellville Rotary (Bellville), 2 Mrs Fourie (Durbanville), 3 Parow Pharmacy (Parow) and 4 Mr Naidoo (Goodwood).

Write the result of this query:

SQL

```
SELECT Item, Qty
FROM tblDonation
WHERE Qty >= 30 AND DonationDate < #2026/06/01#
ORDER BY Qty DESC;
```

Item	Qty

D2 **[5]**

Write the result of this query (it uses both tables):

```sql
SELECT Town, SUM(Qty) AS Total
FROM tblDonor, tblDonation
WHERE tblDonor.DonorID = tblDonation.DonorID
GROUP BY Town
HAVING SUM(Qty) >= 60
ORDER BY SUM(Qty) DESC;
```

Town	Total

D3 **[4]**

Write a query that shows the **donor name** and the **item** of every donation made on or after **1 June 2026**, in order of date.

D4 **[3]**

Write **three** statements: (a) add a donation of 20 Soap from donor 2 on 22 June 2026; (b) double the Qty of every donation from donor 4; (c) delete every donation made before 11 May 2026.

D5 **[4]**

The user types a donor's name in **edtDonor**. **qryFood** is a TADOQuery. Write code that lists in **memOut** the item and quantity of every donation of that donor, in order of date, as *Soap: 60*.

UNIT 7 Hardware and software

Paper 2 tests Grades 10, 11 and 12 together, so this booklet revises all three years, topic by topic. The tag in each title shows the year in which the skill was first taught: [G10], [G11] or [G12]. Watch which tag you drop marks on. One writing rule for the whole booklet: every written answer is a sentence or a short paragraph, never a list of bullets, because a marker credits only as many bullet points as there are marks but reads a paragraph in full. This unit covers devices, memory and storage, performance, the operating system and the programs that translate code.

By the end of this unit I can: (tick each box when you are sure)

- ☐ sort input, output and storage devices
- ☐ compare cache, RAM, ROM, virtual memory, SSD and HDD
- ☐ choose the upgrade that fixes a slow computer
- ☐ explain multitasking, virtual memory and the work of the operating system
- ☐ differentiate between a compiler and an interpreter
- ☐ motivate a computer system for a user

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

7.1 [G10] Fast, small and costly

A computer can keep data in four places: **cache, RAM, an SSD and an HDD**. (a) Write them in order from **fastest to slowest**. (b) Write them in order from **smallest to largest** capacity. (c) In a sentence, say why we do not simply build a PC that has only cache.

LEARN IT

Memory, storage and the system unit

The CPU can only work with data it can reach quickly, so a computer uses a **ladder** of storage: small and fast at the top, big and slow at the bottom.

Part	Job	Keeps data without power?	Remember
Cache (L1, L2, L3)	Copies of recently used instructions and data, right next to the CPU	No (volatile)	Fastest and smallest; more cache means the CPU waits less for RAM
RAM	Working memory for programs and data in use now	No (volatile)	Switch off and it is empty; more RAM lets more programs run
ROM / flash chip	Holds the **BIOS** (or firmware), which runs the **POST** and then loads the OS	Yes	Read-only does not mean "the hard drive"
Virtual memory	Part of the drive that the OS uses as extra RAM when RAM is full	On the drive	Far slower than RAM; too much swapping is **thrashing**
SSD	Flash storage with no moving parts	Yes	Fast, silent, tough, but dearer per GB
HDD	Spinning magnetic platters	Yes	Cheap per GB, but slow, noisy and fragile

On the **motherboard** the CPU talks to the rest over **buses**: the **data bus** carries data and instructions, the **address bus** carries memory addresses and the **control bus** carries signals. A bus is shared; a **point-to-point** link joins exactly two parts (CPU to RAM). **Expansion cards** (graphics, sound, network) plug into PCIe slots. Input devices

send data in (scanner, microphone), output devices present it (printer, projector), a **touchscreen** is both, and storage devices keep it.

LEARN IT

Performance and mobile hardware

Factor	How it changes performance	Typical symptom it fixes
CPU speed (GHz) and number of cores	More instructions per second; more cores handle more jobs at the same moment	Slow calculation and video export
RAM amount and speed	More programs and data stay in fast memory, so less swapping	Freezing with many programs open; drive light always on
Storage type	An SSD loads the OS and programs far faster than an HDD	Long start-up and slow program opening
GPU (graphics card)	Draws graphics so the CPU does not have to	Games and video editing stutter
NIC speed	How fast data enters and leaves the computer on the network	Slow file copying across a LAN

A **smartphone** puts the CPU, GPU, modem and often the memory on one chip, a **system on a chip (SoC)**, which saves space and power and gives off less heat. It has a touchscreen, flash storage and built-in Bluetooth, Wi-Fi, 4G/5G, GPS and NFC. Its limits are the **battery** (faster processing drains it sooner), little room for cooling, a small screen and parts that cannot be upgraded.

LEARN IT

The operating system, utilities and translators

The **operating system (OS)** manages memory (including virtual memory), runs and switches between programs, handles files, talks to devices through **drivers**, controls users and security, and gives you the interface. OS types: **stand-alone** (Windows 11 on a PC), **network/server**, **embedded** (the router, an ATM) and **mobile** (Android). **Utility software** maintains the computer: antivirus, backup, disk clean-up, compression tools. **Multitasking**: one CPU switches so fast between programs that they seem to run together. **Multithreading**: one program is split into threads that progress together. **Multiprocessing**: two or more cores really work at the same instant.

	Compiler	Interpreter
Translates	The whole program, once, before it runs	One line at a time, while it runs
Result	A separate executable (.exe) that runs without the source code	No separate file: the source code and interpreter are needed every time
Speed and errors	Runs fast; all syntax errors are listed before it will run	Runs slower; stops at the line with the error

Licences: *proprietary* (bought, copies limited), *open source* (source code open to change and share), *freeware* (free, closed code) and *shareware* (free trial, then pay).

7.2 [G10] Input, output, both or storage? [6]

Circle the group of each device.

1	A barcode scanner at a spaza till	**Input**	**Output**	**Both**	**Storage**
2	A receipt printer	**Input**	**Output**	**Both**	**Storage**
3	A touchscreen on a self-service kiosk	**Input**	**Output**	**Both**	**Storage**
4	A memory card in a dashcam	**Input**	**Output**	**Both**	**Storage**
5	A projector in the school hall	**Input**	**Output**	**Both**	**Storage**
6	A microphone	**Input**	**Output**	**Both**	**Storage**

7.3 [G10] Match the term [7]

Write the term from the word bank next to each description.
Word bank: BIOS · cache · driver · firmware · POST · ROM · utility

Description	Term
Very fast memory on or beside the CPU that holds recently used instructions	
The start-up program on the motherboard that finds the operating system and loads it	
The check of RAM, keyboard and storage that runs when the PC is switched on	
Memory that keeps its contents when the power is off	
Software stored permanently on a chip inside a device such as a router	
A small program that lets the operating system control a particular printer	
A small program that maintains the computer, such as a backup or disk clean-up tool	

7.4 [G11] Compare: compiler and interpreter [5]

Sipho writes a stock-control program in Delphi and sells it to 40 spaza shops. **Differentiate** between a compiler and an interpreter and say which of them produced the file that the shops run. Answer in a short paragraph.

7.5 [G12] Pick the upgrade [5]

Write the one change that would help most in each case.
Word bank: faster NIC and switch · more RAM · SSD instead of HDD · scan for malware · graphics card (GPU)

Symptom	Best fix
A lab PC takes three minutes to start and every program is slow to open, yet only one program is running	
Everything freezes and the drive light stays on once six browser tabs and a spreadsheet are open	
A school's video-editing PC plays footage smoothly but the preview stutters while effects are drawn	
Pop-up adverts appear and the PC has slowed down although nothing new was installed	
Copying big files between lab PCs takes very long although the PCs themselves are fast	

7.6 [G12] Motivate a computer [5]

The school's media club edits football matches filmed in 4K. **Motivate** the specification of a computer for the club. Refer to the CPU, RAM, graphics and storage, and give a reason for each choice, in a paragraph.

7.7 [G12] Phone versus desktop [4]

Explain why a smartphone uses a system on a chip (SoC) and why its battery limits how fast its processor may run.

UNIT 8 Networks and the internet

From the earphones in your ear to the fibre line under the road, every network is built from the same few ideas: how far it reaches, how it is laid out, which devices join it, which rules (protocols) it follows and how it is kept private. This unit puts Grade 10 networks, Grade 11 protocols and topologies and Grade 12 remote access, SSL and web technologies on one page.

By the end of this unit I can: (tick each box when you are sure)

- ☐ name the network type and compare the topologies
- ☐ state what a switch, router and access point do
- ☐ match protocols to their jobs
- ☐ calculate a download time
- ☐ explain how a VPN, SSL and a digital certificate protect data
- ☐ describe web technologies and SEO

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

8.1 [G11] The cable that was pulled out

A computer lab has ten PCs, each cabled to one central switch. (a) Thabo pulls the cable out of PC 4. What happens to the other nine PCs? (b) During load shedding the switch loses power. What happens to the lab? (c) The teacher says: "Add a second switch and a second link." Which problem is she trying to solve?

LEARN IT

Network types, layouts and devices

Type	Reach	Example
PAN	A few metres around one person	Earbuds, smartwatch and phone (Bluetooth)
HAN	One home	Router, smart TV, phones and a printer in a flat
LAN	One building or campus, usually your own cables	The school's labs and offices
WAN	Cities, countries; uses lines leased from others	A bank's branches; the Internet is the biggest WAN

Topology	Layout	Strength	Weakness
Star	Every device joined to a central switch	A cable fault affects one device; easy to add devices	If the switch fails, everyone fails
Bus	All devices share one backbone cable	Cheap and simple	A break in the backbone stops all; slows down as devices are added
Ring	Each device joined to two neighbours in a loop	Orderly, few collisions	One break can stop the whole ring
Mesh	Devices linked to several others	Very reliable: data finds another route	Many cables, so expensive

A **NIC** connects a device to the network. A **switch** joins the devices of **one** LAN and sends data only to the device it is meant for. A **router** joins **different** networks (your LAN and the Internet) and chooses the route using IP addresses. A **wireless access point** lets Wi-Fi devices join the wired network. A **firewall** filters traffic by rules; a **modem** converts between the line signal and the computer's data. In a **client-server** network a powerful server holds shared files, users and security and clients ask it for services: easy to manage and secure, but costly, and the

server is a single point of failure. In a **peer-to-peer** network every computer is equal and shares directly: cheap and simple, but no central control and weak security. A **thin client** does most of its processing on the server; a **thick client** does it locally.

LEARN IT

Connections and protocols

Connection	Strength	Weakness
Fibre	Fastest, very stable, low delay	Needs a cable to the building
ADSL (copper phone line)	Widely available	Slower; speed drops with distance
4G / 5G	No cable needed; 5G has very low delay	Expensive capped data; signal varies
Satellite	Works almost anywhere	High delay (latency); costly
Wi-Fi / **Bluetooth** / **NFC**	Wireless inside a building / about 10 m / a few cm (tap to pay)	Short range; Wi-Fi is easier to attack

Protocol	Job
HTTP / **HTTPS**	Web pages: HTTP as plain text, HTTPS encrypted with SSL/TLS
SMTP	**Sends** e-mail to and between mail servers
POP3	**Downloads** e-mail to one device to read offline
IMAP	Keeps e-mail **on the server**, so every device sees the same mailbox
FTP	Moves large files between a computer and a server
VoIP	Voice and video calls as packets over the Internet

Bandwidth is how much data fits through per second; **latency** is the delay. Speeds are in megabits (Mb) per second but files are in megabytes: **1 byte = 8 bits**, so divide by 8 to compare. Time = file size in megabits ÷ speed in Mbps.

LEARN IT

Remote access, SSL and the web

A **VPN** builds an encrypted tunnel across the public Internet, so a person at home reaches the private network as if she were in the office and anyone who intercepts the data sees only scrambled text. **SSL/TLS** secures a browser-to-server connection (HTTPS and a padlock). The server shows its **digital certificate**, signed by a trusted authority, proving the site is genuine. The browser locks a secret **session key** with the server's **public key**; only the server's **private key** can unlock it, and both then encrypt the data with the session key. A **static** web page is the same for everyone; a **dynamic** page is built for you from a database.

Technology	Where it runs	Job
HTML	Browser	Content and structure of the page
CSS	Browser	Colours, fonts and layout
JavaScript	Browser (client-side)	Reacts at once, for example checks a form before it is sent
PHP	Web server (server-side)	Builds pages and talks to the database
AJAX	Browser asks server in the background	Updates part of a page without reloading it
Cookie	Stored on your device	Lets a site remember you (login, basket) and lets advertisers follow you

A **keyword search** matches the typed words, a **semantic search** works out the meaning and intention, and a **mediated search** looks only in a controlled collection (a library database). **SEO** improves a site's rank by using the words people really search for, loading fast, working on phones and earning links from other sites.

8.2 [G10] PAN, HAN, LAN or WAN? [5]

Circle the network type.

1	Wireless earbuds linked to a phone	**PAN** **HAN** **LAN** **WAN**
2	The smart TV, phones and router in one flat	**PAN** **HAN** **LAN** **WAN**
3	The labs and offices of one school, joined by a switch	**PAN** **HAN** **LAN** **WAN**
4	A bank's link between its Cape Town and Durban offices	**PAN** **HAN** **LAN** **WAN**
5	The Internet	**PAN** **HAN** **LAN** **WAN**

8.3 [G10] Match the protocol [7]

Write the protocol from the word bank next to each job.
Word bank: FTP · HTTP · HTTPS · IMAP · POP3 · SMTP · VoIP

Job	Protocol
Sends an e-mail from your device to the mail server	
Downloads your e-mail so you can read it offline on one device	
Lets your phone and laptop show the same mailbox because messages stay on the server	
Moves a large file between a computer and a server	
Carries web pages as plain, unencrypted text	
Carries web pages encrypted with SSL/TLS	
Carries a voice call as small packets over the Internet	

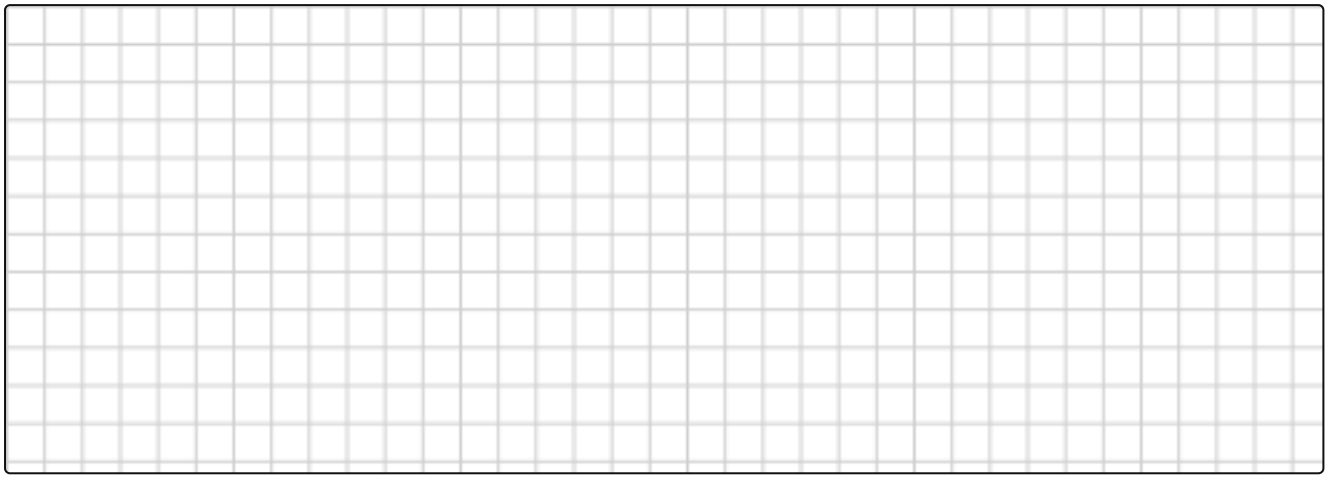
8.4 [G11] How long is the download? [4]

A school downloads a **750 MB** update (1 byte = 8 bits, so the file is 6 000 megabits). Complete the time for each connection, assuming the full speed is used.

Connection	Speed (Mbps)	Time in seconds	Time in minutes
Fibre	100		
ADSL	20		
4G	10		
Satellite	2		

8.5 [G11] Draw the office network [5]

A taxi association office has **6 PCs**, a **printer**, a **file server** and a manager who uses a **tablet** over Wi-Fi. A fibre line arrives from the ISP. Draw a **star** network showing every device, the devices that connect them and where the **firewall** sits. Label everything.



8.6 [G12] Name the web technology [7]

Write the term from the word bank.
Word bank: AJAX · cookie · CSS · HTML · JavaScript · PHP · SEO

Description	Term
Describes the content and structure of a page: headings, paragraphs and links	
Controls the colours, fonts and layout of a page	
Runs in the browser and checks a form before it is sent	
Runs on the web server, builds the page and talks to the database	
Fetches new data in the background so only part of the page refreshes	
A small file stored on your device so a site can remember you	
Using the words people search for and loading fast so that a website ranks higher in search results	

8.7 [G12] Explain: SSL and the padlock [5]

Lindiwe buys airtime stock on a wholesaler's website. The address starts with https and shows a padlock. **Explain** how the digital certificate and the public and private keys protect her card number.

8.8 [G12] Motivate: fibre or a 4G router? **[5]**

A township internet cafe wants fast video calls for customers. Fibre is available and so is a 4G router. **Motivate** which should be the main link and which the backup, and say how load shedding should be handled. Answer in a paragraph.

UNIT 9 Data and databases

A database is only as good as the data in it and the design under it. This unit revises how data becomes information, how we keep it correct, how tables are linked and designed on paper (normalisation), and how big organisations store and mine data.

By the end of this unit I can: (tick each box when you are sure)

- ☐ tell data from information and explain GIGO
- ☐ choose the right validation check and work out a check digit
- ☐ read and draw an ERD with keys and a link table
- ☐ explain referential integrity and the three anomalies
- ☐ normalise a table to 3NF on paper
- ☐ compare a database and a data warehouse and explain data mining

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

9.1 [G10] Data or information?

The hall thermometer is read every hour: **27, 31, 29, 35** (°C). (a) Which of these is **data** and which is **information**: "27, 31, 29, 35" or "The hall averaged 30.5 °C and was above 30 °C in the afternoon"? (b) What had to be done to turn one into the other? (c) The 31 is typed as **13** by mistake. What average would be reported, and what do we call this problem?

LEARN IT

Data quality, validation and collection

Data is raw facts without meaning; **information** is data that has been processed and put in context. A **DBMS** stores data and controls access, security, backups and multi-user use. **GIGO:** wrong data in means wrong information out. **Validation** is done by the program and checks that data is *reasonable*; **verification** checks that it is the *same as the source* (typing it twice, proofreading against the form). Validation cannot see a mark of 89 typed instead of 98.

Check	Tests	Example
Presence	Something was entered	Surname may not be empty
Range	Between a minimum and a maximum	Mark 0 to 40
Length	Number of characters	Cell number has 10 digits
Type	Kind of data	Age holds digits only
Format	Pattern	E-mail has @ and a dot after it
List	One of the allowed values	Province chosen from nine
Check digit	Last digit is calculated from the others, so a typing slip is noticed	Barcodes, ISBNs, ID numbers

Collecting data: web forms, **RFID** tags read by radio (e-toll, stock), **sensors**, **cookies**, **transaction tracking** (every card swipe) and mobile apps (location). **Metadata** is data about data, such as the date, size and GPS position stored in a photo file.

LEARN IT

Keys, relationships and integrity

A **primary key (PK)** identifies each record: never empty, never repeated. A **foreign key (FK)** is a field that holds another table's primary key and so creates the link; it may repeat and may be empty. A **composite key** uses two or more fields together.

Relationship	Meaning	How it is built
One-to-one (1:1)	One record matches one record	The same key in both tables
One-to-many (1:M)	One record matches many	Put the PK of the 'one' table into the 'many' table as an FK
Many-to-many (M:M)	Many match many	**Cannot be stored directly**: add a **link table** that holds both PKs as FKs, giving two 1:M relationships

Referential integrity: every foreign key must match an existing primary key. It stops you adding a record that points at nothing (an **orphan**) and stops you deleting a record that others still use. **Logical integrity** keeps data correct and consistent (validation, referential integrity, **record locking**, which stops two users editing the same record at once); **physical integrity** protects it from damage (UPS, RAID, backups).

LEARN IT

Normalisation, anomalies and big data stores

Repeating the same fact in many rows (**redundancy**) causes three **anomalies**: **insertion** (you cannot record one fact without another), **deletion** (deleting one fact wipes out another) and **modification** (a change must be made in many rows, and missing one makes the data disagree with itself). **Normalisation** removes them by splitting a table. **1NF:** one value per cell, no repeating columns, and a primary key. **2NF:** in 1NF, and every other field depends on the *whole* composite key (move a field that depends on part of the key to its own table). **3NF:** in 2NF, and no non-key field depends on another non-key field (move that pair out).

	Database	Data warehouse
Purpose	Day-to-day transactions	Analysis and reports over many years
Data	Current	Historical, collected from many databases, cleaned first
Used by	Staff and programs that insert, update and delete	Analysts and management, who mostly read

Data mining searches very large data sets for hidden patterns, such as which products are bought together. A **centralised** database sits on one server (simple, but one point of failure); a **distributed** one is spread across several servers or sites (faster for distant users, but copies must be kept in step).

9.2 [G11] Choose the check [6]

Write the validation check from the word bank.
Word bank: format · length · list · presence · range · type

Rule	Check
The surname box may not be left empty	
A test mark is out of 40, so it must be 0 to 40	
A South African cell number must have exactly 10 digits	
The age box may contain only digits	
An e-mail address must contain an @ with a dot after it	
The province must be picked from the nine allowed names	

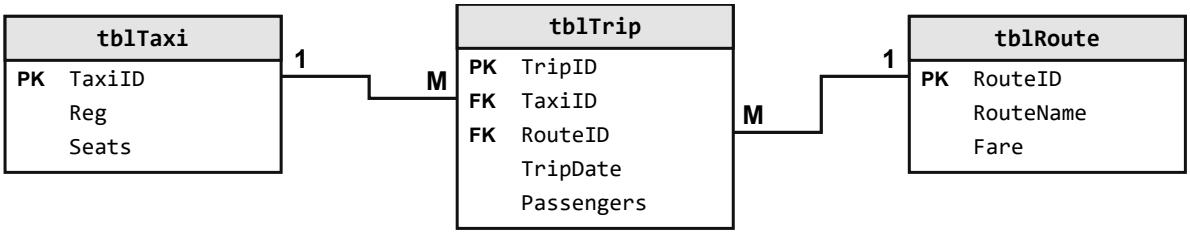
9.3 [G11] Work out the check digit [7]

A product code has 4 digits plus a check digit. **Multiply the digits by 3, 1, 3, 1** in turn and add. The check digit is the number needed to reach the **next multiple of 10** (0 if the sum already is one). Example: code 4276 gives 3×4 + 2 + 3×7 + 6 = 41, so the check digit is 9 and the full code is 42769. Complete the first three rows. For the last two, work out the correct check digit and say whether the code is valid; leave the last column of the first three blank.

Code	Weighted sum	Correct check digit	Valid?
5083			
7219			
4650			
72195			
46508			

9.4 [G11] Read the ERD [5]

A taxi association stores its trips in the database below. Answer from the diagram.



Question	Answer
The primary key of tblTrip	
The two foreign keys in tblTrip	
The relationship between tblTaxi and tblTrip (one-to-many?)	
The many-to-many relationship that tblTrip resolves	
Explain one thing that referential integrity stops in tblTrip	

9.5 [G11] Draw the ERD: donors and events [5]

A school records donors and fundraising events. A donor can give to **many** events and an event receives gifts from **many** donors. Draw an ERD with **tblDonor**, **tblEvent** and a link table **tblGift**. Write the fields, mark every PK and FK and put **1** or **M** at the right ends.

9.6 [G12] One value per cell (1NF) [6]

The table below is **not** in 1NF because the Club column holds several values. Rewrite it in 1NF: give each club its own row. Then the primary key must be LearnerNo and Club together.

LearnerNo	Name	Club
1021	Aphiwe Dlamini	Chess, Debating
1034	Ruan Botha	Choir
1047	Sana Patel	Chess, Choir, Robotics

LearnerNo	Name	Club (one per row)
1021	Aphiwe Dlamini	
1021	Aphiwe Dlamini	
1034	Ruan Botha	
1047	Sana Patel	
1047	Sana Patel	
1047	Sana Patel	

9.7 [G12] Whose field is it? [6]

The stock-order table below has the composite primary key **OrderNo + ItemCode**. For each other field say what it depends on and which problem (if any) it causes.

OrderNo	ItemCode	ItemName	ItemPrice	Qty	OrderDate	SupplierID	SupplierName
801	B12	Bread	R18.50	40	02 Mar	S1	Bellville Cash and Carry
801	M07	Maize meal	R52.00	12	02 Mar	S1	Bellville Cash and Carry
802	B12	Bread	R18.50	25	09 Mar	S2	Cape Wholesalers
802	D03	Milk 2 L	R34.90	18	09 Mar	S2	Cape Wholesalers
803	M07	Maize meal	R52.00	10	16 Mar	S1	Bellville Cash and Carry

Field	Depends on	Problem
Qty		
ItemName		
ItemPrice		
OrderDate		
SupplierID		
SupplierName		

9.8 [G12] Write the 3NF tables [7]

Split the stock-order table into 3NF. Write each table with its fields and mark the primary keys (PK) and foreign keys (FK).

9.9 [G12] Which anomaly? [3]

Circle the anomaly shown in each case. Use the stock-order table above.

1	A new supplier, Karoo Foods, cannot be stored until someone orders from it.	**Insertion** **Deletion** **Modification**
2	Order 802 is cancelled and its rows deleted; Cape Wholesalers and the price of milk vanish with it.	**Insertion** **Deletion** **Modification**
3	The bread price rises, but only one of its 2 rows is changed, so the table shows two prices.	**Insertion** **Deletion** **Modification**

UNIT 10 Number systems and data representation

Everything on a computer, from your name to a video of the match, is stored as patterns of 0s and 1s. This unit revises counting in binary and hexadecimal, how characters and pictures are stored, how big files are, and what compression gives up.

By the end of this unit I can: (tick each box when you are sure)

- ☐ convert between decimal, binary and hexadecimal
- ☐ use ASCII and explain why Unicode is needed
- ☐ convert between storage units
- ☐ calculate the size of a picture and its compressed size
- ☐ tell lossy from lossless compression

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

10.1 [G10] Eight switches

A spaza owner has a row of light switches. With **1** switch there are 2 patterns (off or on). How many different patterns can **2** switches make? And **3**? Spot the rule, then work out how many patterns **8** switches make and the largest number they can show if each switch is worth 1, 2, 4, 8, 16, 32, 64 and 128 when it is on.

LEARN IT

Counting like a computer

Place values in binary double each step: **128 64 32 16 8 4 2 1**. **Binary to decimal:** add the place values that have a 1 (00101101 = 32 + 8 + 4 + 1 = 45). **Decimal to binary:** divide by 2 again and again and read the remainders from the bottom up, or subtract place values from the left. **Hexadecimal** (base 16) uses 0 to 9 and A (10) to F (15), and one hex digit is exactly **4 bits**, so split a byte into two nibbles: 1010 1101 = A D = AD. To convert hex to decimal use 16s: AD = 10×16 + 13 = 173. n bits give $\mathbf{2^n}$ patterns.

Text and storage	
ASCII	7-bit code (128 characters), usually stored in a byte; English letters, digits and symbols. A = 65, a = 97, 0 = 48, space = 32.
Unicode	Uses more bits per character (UTF-8, UTF-16, UTF-32) so it can hold every script (isiZulu, Chinese, Arabic) and emoji; the first 128 codes are the same as ASCII.
Units	1 byte = 8 bits; 1 KB = 1 024 bytes; 1 MB = 1 024 KB; 1 GB = 1 024 MB; 1 TB = 1 024 GB. Multiply to go down a unit, divide to go up.

LEARN IT

Pictures, files and compression

A true-colour (24-bit) picture needs **3 bytes per pixel**, so **size in bytes = width × height × 3**. Divide by 1 024 for KB and again for MB. A **compression ratio** of 20:1 means the compressed file is one twentieth of the original; **saving (%) = (original − compressed) ÷ original × 100**. **Lossless** compression (ZIP, PNG) lets you rebuild the original exactly, so use it for documents, logos and medical images. **Lossy** compression (JPEG, MP3, MPEG-4) throws away detail people hardly notice, so the file is much smaller but the original cannot be rebuilt; more compression means lower quality.

10.2 [G10] Decimal to binary and hex [6]

Write each number as an **8-bit binary** number and as **hexadecimal**.

Decimal	8-bit binary	Hexadecimal
37		
90		
173		

10.3 [G10] Back to decimal [4]

Convert each number to decimal. Show your working in the margin.

Number	Decimal
11010110 (binary)	
01101001 (binary)	
7C (hex)	
B3 (hex)	

10.4 [G10] ASCII [7]

Use A = 65, a = 97 and 0 = 48 (the codes run in order). Complete the table; in the last row decode the four bytes to a word.

Character or bytes	ASCII decimal	Binary or word
T		
x		
7		
01010100 01000001 01011000 01001001		

10.5 [G11] Storage sums [5]

Use 1 KB = 1 024 bytes, 1 MB = 1 024 KB and 1 GB = 1 024 MB.

Question	Answer
(a) Convert 3.5 GB to MB	
(b) How many 4 MB photos fit on a 2 GB flash drive?	
(c) Convert 6 144 KB to MB	
(d) How many bytes are in 8 KB?	
(e) A 500 GB drive holds 380 GB of data. What percentage of the drive is free?	

10.6 [G11] Picture size and saving [6]

A true-colour photo is **1 280 × 720 pixels** (3 bytes per pixel). Work out its size and its size after JPEG compression at **20:1**.

Working

```
pixels := width * height
bytes  := pixels * 3
KB     := bytes / 1024
MB     := KB / 1024
JPEG   := bytes / 20
saving := (bytes - JPEG) / bytes * 100
```

Step	Working	Answer
1	1 280 × 720	
2	pixels × 3	
3	bytes ÷ 1 024	
4	KB ÷ 1 024 (2 decimals)	
5	bytes ÷ 20	
6	saving as a percentage	

10.7 [G11] Motivate: pictures for the school website [5]

The school website will show 40 action photos of sport and the school badge (flat colours). **Motivate** the compression you would use for each, and explain why in a paragraph.

UNIT 11 Security, crime and society

Technology brings risks and changes how we live and work. This unit revises the malware and attacks that threaten a system, the safeguards that stop them, cybercrime and the law, and the social effects of the connected world, with paragraph practice for the long questions.

By the end of this unit I can: (tick each box when you are sure)

- ☐ name the malware or attack from a description
- ☐ match a safeguard to a threat and plan a backup
- ☐ write a paragraph on a cybercrime, its effects and its safeguards
- ☐ explain what POPIA requires
- ☐ explain big data, IoT, AI, VR, AR, cloud and 4IR with their benefits and risks
- ☐ discuss the digital divide and green computing

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

11.1 [G10] The office PC audit

Mrs Naidoo, the school secretary, has a PC with her password on a sticky note, learners' marks on it that were never copied anywhere, a flash drive that goes home with her every day, and no screen lock when she walks away. Write **four** different things that could go wrong, and next to each the fix.

LEARN IT

Malware, attacks and safeguards

Malware	How it works
Virus	Attaches to a file and spreads when a person opens that file
Worm	Spreads **by itself** across a network, no user action needed
Trojan	Looks like useful software; does harm once you run it
Rootkit	Hides deep in the OS and gives an attacker hidden control
Ransomware	Encrypts your files and demands payment to unlock them
Spyware / **adware**	Secretly records what you do / shows unwanted adverts
Botnet	Many infected computers (zombies) controlled in secret, often to send spam or attack a site

Attack	How it works
Phishing	A fake e-mail or SMS tricks you into giving your details; **you** click the link
Pharming	Sends you to a fake website even if you typed the right address
Spoofing	Forges the sender's address or identity
DDoS	Floods a server with traffic from many computers so real users cannot get in
SQL injection	Malicious SQL typed into a web form to read or change the database
Social engineering	Tricks a person, not a machine, into revealing information

Safeguards: antivirus and updates, a firewall, strong passwords with **MFA**, access rights (least privilege), encryption, input validation (against SQL injection), user training, a **UPS** and surge protection for load shedding, and **backups**. The **3-2-1 rule**: three copies, on two kinds of media, one of them off-site or offline (so ransomware or a fire cannot reach it).

LEARN IT

Cybercrime, criminals and the law

Criminals: a *black-hat hacker* breaks in for harm or gain (a *white-hat* hacker does it with permission to find weaknesses); a *cracker* breaks software protection; a *script kiddie* uses tools made by others; an *insider* misuses the access he already has. **Other cybercrimes:** identity theft, online fraud, cyberbullying, piracy, bandwidth theft, piggybacking on someone's Wi-Fi and cyber-extortion. **Effects:** a person loses money, privacy and peace of mind; a business loses money, customers' trust, trade secrets and time while systems are down, and may be fined. **South African law:** the **Cybercrimes Act** makes hacking, ransomware, fraud and malicious messages crimes; **POPIA** says an organisation may collect personal information only for a stated reason and with consent, must keep it accurate and secure, may keep it no longer than needed, and must report a breach; the **ECT Act** covers electronic transactions.

LEARN IT

Technology and society

Big Data is data too large, too fast or too varied for ordinary tools: **volume**, **velocity**, **variety**. The **Internet of Things** is physical objects with sensors that send and receive data automatically. **4IR** merges the physical, digital and biological worlds (AI, robots, IoT, Big Data): routine jobs may disappear, new ones appear and workers must retrain. **AI and machine learning** let computers learn from data to decide or predict; the risks are bias in the data, lost jobs and decisions nobody can explain. **Virtual reality (VR)** puts you inside a computer-made world with a headset; **augmented reality (AR)** adds computer images to the real world you still see. **Cloud computing** rents storage and software over the Internet: pay only for what you use, reach your files anywhere, backups and updates handled for you; but you depend on the connection (and data costs and load shedding), your data sits on someone else's servers and privacy is a worry. A **blockchain** is a chain of records, each holding the hash of the block before, copied on many computers, so an old record cannot be changed without notice. The **digital divide** separates people who have devices, data and skills from those who do not (cost, rural areas, skills). **Green computing** saves energy and cuts e-waste: power-saving settings, switching off, recycling, longer-lasting devices, cloud sharing. Your **digital footprint** is the trail of data you leave: *active* (posts, forms) or *passive* (cookies, location logs).

11.2 [G11] Name the malware [8]

Write the type of malware from the word bank.
Word bank: adware · botnet · ransomware · rootkit · spyware · Trojan · virus · worm

Description	Malware
Spreads across the school network by itself, with nobody opening anything	
Looks like a free game, but gives a criminal access once it is run	
Locks every file on the office PC and demands R5 000 for the key	
Records every key you press, including passwords	
Shows pop-up adverts you never asked for	
Attaches itself to a document and spreads when the document is opened	
Hides deep inside the OS so that antivirus does not notice it	
A group of infected PCs, secretly controlled, used to flood a website	

11.3 [G11] Name the attack [6]

Write the attack from the word bank.
Word bank: DDoS · pharming · phishing · social engineering · spoofing · SQL injection

What happened	Attack
A fake bank e-mail asks the customer to click a link and type her PIN	
The customer types the bank's correct address but lands on a copy of the site	
A criminal types code into the login box that makes the database show every user	
Thousands of computers request an online shop's page at the same moment, so it crashes	
A caller pretends to be from IT and asks the clerk for her password	
A message appears to come from the principal's e-mail address but does not	

11.4 [G10] Threat and safeguard [6]

Write the safeguard that best deals with each threat.
Word bank: access rights · antivirus · backup kept off-site · encryption · firewall · UPS

Threat	Safeguard
A power cut during load shedding switches the server off in the middle of saving	
A thief steals a laptop that holds learners' personal details	
A flood destroys the office, the PC and the backup disk beside it	
An attacker outside the school tries to connect to the LAN	
A learner opens marks that only teachers should see	
A downloaded game contains a virus	

11.5 [G11] Plan the backup [5]

A spaza owner keeps all her sales and stock records on one laptop at the shop. **Explain** the 3-2-1 backup rule and why one copy must be kept off-site or offline, using a theft or a ransomware attack as your example.

11.6 [G12] Cybercrime: the fake SMS [7]

A fake SMS tells taxi drivers that their licences are suspended and that they must click a link and pay a fine. **Identify** the cybercrime, **explain** how it works, **discuss** two effects on a driver and give two safeguards. Write a paragraph.

11.7 [G12] POPIA [4]

The school keeps learners' ID numbers, addresses and marks. **Explain** two things that POPIA requires the school to do with this information.

11.8 [G12] Match the technology [6]

Write the technology from the word bank.
Word bank: AI and machine learning · augmented reality · blockchain · cloud computing · Internet of Things · virtual reality

Description	Technology
Rents storage and software over the Internet and bills only for what is used	
A headset puts the user inside a computer-made world	
A phone camera shows a car's service history on top of the real car	
Computers learn from past data to recognise faces or predict faults	
A shared chain of records, each linked to the one before, copied on many computers	
Objects with sensors, such as a prepaid meter, that send data automatically	

11.9 [G11] Discuss: the digital divide [4]

Explain what the digital divide is, give two reasons for it in South Africa and say one way to reduce it.

11.10 [G10] Green computing **[3]**

Explain two ways in which a school can practise green computing and why each helps the environment.

UNIT 12 Exam answering skills

Marks are lost in the exam room for avoidable reasons: a command word ignored, a mark allocation not counted, an answer that names things but never explains them. This unit trains the habits that protect marks before you sit the mini-exam.

By the end of this unit I can: (tick each box when you are sure)

- ☐ underline the command word and count the points a question needs
- ☐ turn a list of words into a paragraph with a reason in every sentence
- ☐ plan a long answer in the margin
- ☐ judge an answer against a marking guideline
- ☐ work out how many minutes each mark deserves

One thing I want to ask my teacher: _____

LEARN IT

Read, count, plan, write

Read the whole question and underline the command word. **Count** the marks: usually one separate, correct point per mark, so [3] means three points. **Plan** the points in the margin. **Write** so that every point has the shape *point → because → so/for example*, using the names in the scenario (the spaza shop, the taxi association, the school). **Check** that you made as many points as there are marks, and never leave a question blank. A point that is named but not explained usually earns half or nothing.

Command word	What to do
Name / State / List / Identify	Give the answer only: a word or short phrase for each mark.
Define / Describe	Say what it is or what it does, in a sentence.
Explain	Say what and why or how: a point and its reason for each mark.
Differentiate / Compare	Show how two things differ with both things in every sentence.
Discuss	Give both sides (advantage and disadvantage, benefit and risk) and explain each.
Motivate / Justify / Recommend	Choose, then give reasons; the word *because* must appear.
Calculate	Formula, working and answer with its unit.

LEARN IT

A model paragraph, mark by mark

Discuss one advantage and one disadvantage of paying taxi fares with a card instead of cash. [4]

Sentence of the answer	Mark	Why it earns the mark
Card payments are an advantage because the driver does not have to carry or count cash,	1	Advantage named
so he is less likely to be robbed and the fare is recorded automatically.	1	Advantage explained
A disadvantage is that the system needs a working network connection and electricity,	1	Disadvantage named
so during load shedding or a signal failure passengers cannot pay and the driver loses income.	1	Disadvantage explained with a result

Notice that each pair of sentences is a **point and its reason**, in the scenario's words. The same four ideas as bullets ("less cash – recorded – needs network – load shedding") show no reasoning, and a marker may stop reading after the marks available.

12.1 [G12] Plan, then write [4]

Discuss one advantage and one disadvantage of a municipality using an AI chatbot to answer residents' questions about accounts. [4]
Write your plan in the margin first (a few words for each point), then your paragraph underneath.

12.2 [G12] Mark it, then fix it [3]

Explain why shops should keep the software on their tills up to date. [3]
Learner's answer: "Updates are important. Software must be updated so it is up to date. It is good for the shop."
The marking guideline gives one mark each for: (1) updates repair security weaknesses; (2) criminals use known weaknesses to infect or break in; (3) updates also fix faults and add features, so the tills run reliably. (a) How many marks does the learner earn? Why? (b) Rewrite the answer so that it earns all three.

12.3 [G12] Time per mark [6]

The NSC Paper 2 has 150 marks and lasts 180 minutes, so each mark gets _____ minutes and a 6-mark question about _____ minutes. If you keep 10 minutes at the end for checking, you have _____ minutes for the questions, which is _____ minutes per mark. The mini-exam in this booklet has _____ marks in 120 minutes, so you have about _____ minutes per mark.

PAPER 2 MINI-EXAM

Theory, Grades 10 to 12

Time: 120 minutes **Total:** 77 marks **Mark:** _____ / 77

No notes, no computer. Answer in full sentences or short paragraphs, never bullet points. Read each question twice and count the marks.

SECTION A — Short questions [21]

A1 [G10] Circle the correct word [5]

Circle the correct answer in each case.

1	The BIOS is stored in	**RAM** **ROM**
2	Which has no moving parts?	**HDD** **SSD**
3	Which translator converts the whole program before it runs?	**Compiler** **Interpreter**
4	Which malware spreads across a network by itself?	**Virus** **Worm**
5	Which e-mail protocol keeps the messages on the server?	**POP3** **IMAP**

A2 [G10] Number systems [4]

Answer in the table.

Question	Answer
(a) Write decimal 59 as an 8-bit binary number	
(b) Convert binary 10110100 to decimal	
(c) Convert decimal 203 to hexadecimal	
(d) Convert hexadecimal 3D to decimal	

A3 [G11] Which validation check? [4]

Write the check from the word bank.
Word bank: check digit · length check · list check · range check

Rule	Check
A learner number must have exactly 6 digits	
A learner's age must be from 5 to 19	
The grade is chosen from a drop-down showing 8, 9, 10, 11 and 12	
The last digit of a product code is worked out from the other digits	

A4 [G11] Picture size [4]

A true-colour photo is **640 × 480 pixels** at 3 bytes per pixel (1 KB = 1 024 bytes, 1 MB = 1 024 KB).

Question	Answer
(a) Number of pixels	
(b) File size in bytes	
(c) File size in KB	
(d) Size in KB after JPEG compression at 10:1	

A5 [G12] Name the term [4]

Write the term from the word bank.
Word bank: botnet · DDoS attack · metadata · record locking · SoC

Description	Term
One chip that holds the CPU, graphics, modem and memory of a phone	
Data that describes other data, such as the date and GPS position stored in a photo	
Stops two users from editing the same record at the same moment	
Infected computers secretly controlled by a criminal	

SECTION B — Hardware, software and networks [16]

B1 [G11] Slow library PCs [4]

The twelve PCs in the Siyakhula Community Library freeze when four programs are open, and the hard-drive light stays on. **Explain** what virtual memory is, why it causes the freezing and name one upgrade that would solve the problem.

B2 [G10] Which protocol? [4]

Write the protocol from the word bank.
Word bank: FTP · HTTPS · IMAP · POP3 · SMTP

Job	Protocol
Sends an e-mail from the librarian's PC to the mail server	
Lets her phone and PC show the same mailbox	
Carries the library's online payment page with the data encrypted	
Uploads a large file of new books to the supplier's server	

B3 [G10] Why a server? [4]

The library has 12 PCs for the public. The librarian wants a client-server network and not a peer-to-peer network. **Explain** two reasons for her choice.

B4 [G12] Working from home [4]

The librarian wants to open the library's catalogue server from home. **Explain** how a VPN protects the data on the way, and give one remaining risk of remote access.

SECTION C — Data and databases [23]

C1 [G11] Draw the ERD [5]

A clinic keeps a record of every visit. A patient can have **many** visits, each visit is with **one** doctor, and a doctor sees **many** visits. Draw an ERD with **tblPatient**, **tblDoctor** and **tblVisit**. Show the fields, mark each PK and FK and put 1 or M at the correct ends.

C2 [G12] Whose field is it? [4]

The clinic's old table has the composite key **PatientID + VisitDate**. For each field below, say what it depends on and which problem (if any) it causes.

PatientID	VisitDate	PatientName	DoctorID	DoctorName	Fee
2201	05 Aug	Nomsa Zungu	D1	Dr Pillay	R280.00
2201	19 Aug	Nomsa Zungu	D2	Dr Smit	R350.00
2214	05 Aug	Kabelo Mosia	D2	Dr Smit	R320.00
2230	12 Aug	Anele Cele	D2	Dr Smit	R320.00

Field	Depends on	Problem
PatientName		
DoctorID		
DoctorName		
Fee		

C3 [G12] Which anomaly? [3]

Circle the anomaly shown, using the clinic table in the question above.

1 Dr Smit's name is corrected in one of her three rows only, so it is spelled two ways. **Insertion** **Deletion** **Modification**

2 A new doctor cannot be saved until a patient has a visit with her. **Insertion** **Deletion** **Modification**

3 Anele Cele's only visit is deleted and her name disappears from the database. **Insertion** **Deletion** **Modification**

C4 [G12] Normalise to 3NF [6]

Write the clinic table in 3NF. Show each table with its fields and mark the primary keys (PK) and foreign keys (FK).

C5 [G12] Warehouse and mining [5]

A group of 30 clinics keeps a separate database at each clinic. Management wants to study ten years of visits to see in which months flu visits peak. **Explain** what store they should build and how it differs from the clinics' databases, and name the process of finding the patterns.

SECTION D — Security, crime and society [17]

D1 [G12] Cybercrime: the municipal breach [6]

Hackers broke into a municipality's billing database and published residents' ID numbers, addresses and account details. **Discuss** the effects of the breach on the residents and on the municipality, and give two safeguards the municipality should have used. Answer in a paragraph.

D2 [G11] Discuss: IoT street lights [4]

A municipality fits **IoT sensors** to its street lights. **Discuss** one benefit and one risk of the system.

D3 [G12] Cloud for a small office [4]

A small municipal office with no IT staff considers moving its documents to the cloud. **Explain** one advantage and one disadvantage of this decision.

D4 **[G11] The digital divide** **[3]**

The municipality moves all services online. **Explain** what the digital divide is and how this decision could widen it.

UNIT 13 The final stretch

Revision only works when it is planned. Use this page to decide how you will spend the days before each paper, and go back to your self-audit to see how far you have come.

By the end of this unit I can: (tick each box when you are sure)

- ☐ re-do the self-audit and compare it with my first one
- ☐ plan my last days before each paper
- ☐ decide what I will do the evening before the exam

One thing I want to ask my teacher: _____

13.1 My exam-week plan

Plan the last days before each paper. Write when you will do your final run-through, which topics you will go over once more, what you will pack for the exam room, and what you will do the evening before to be rested.